

Tumor-Immune Active Matter

Mathematical and physical model summary

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Code: `src/{sim, fields, interactions, sweep, render, style}.py`
Plan: `docs/PLAN.md` · Decisions: `docs/DECISIONS.md`
LaTeX source: `docs/model.tex` · Markdown: `docs/model.md`

1. Setup

Two species of self-propelled particles + two scalar concentration fields, in a 2D periodic square box of side L .

State variables

- $r_i(t), \theta_i(t)$ – position and heading of tumor cell i , $i = 1, \dots, N_T(t)$
- $R_j(t), \phi_j(t)$ – position and heading of T-cell j , $j = 1, \dots, N_I(t)$
- $c_a(x, t)$ – attractant concentration (long-range)
- $c_s(x, t)$ – suppressant concentration (short-range)
- $\rho_T(x, t)$ – tumor density on the field grid, computed by CIC deposit

Non-conserved dynamics

Tumor cells are born by proliferation (a source term most active-matter models lack) and removed by T-cell-mediated killing. T cells are conserved on the simulation timescale (no proliferation, no natural death).

Theoretical neighborhood

- Toner-Tu hydrodynamics of polar active matter (no proliferation).
- Joanny-Prost-Ranft growing active matter (proliferation as non-conserved source).
- Keller-Segel chemotaxis (the attractant loop is the classical instability).

The novel content is the combination of long-range positive coupling via c_a and short-range negative coupling via c_s , from the SAME tumor source. That asymmetry generates the bistable phase boundary which maps onto the immune-excluded tumor histology.

2. Tumor-cell dynamics

Tumor cells obey overdamped Langevin self-propulsion with soft pairwise repulsion and rotational diffusion. The position equation:

$$\frac{d\mathbf{r}_i}{dt} = v_T \mathbf{n}_i + \sum_{j \neq i} \mathbf{F}_{\text{rep}}(\mathbf{r}_i - \mathbf{r}_j) + \sum_k \mathbf{F}_{\text{rep}}(\mathbf{r}_i - \mathbf{R}_k) + \sqrt{2D_T} \boldsymbol{\eta}_i(t)$$

with $\mathbf{n}_i = (\cos \theta_i, \sin \theta_i)$. Heading evolves by:

$$\frac{d\theta_i}{dt} = \sqrt{2D_R^T} \xi_i(t)$$

η_i and ξ_i are independent unit-variance Gaussian white noises. The repulsion is harmonic with cutoff at one cell diameter σ :

$$\mathbf{F}_{\text{rep}}(\mathbf{d}) = k_{\text{rep}} \max(0, \sigma - \|\mathbf{d}\|) \hat{\mathbf{d}}$$

Proliferation

Each step, each alive tumor cell tries to divide with probability p_{div} , subject to a local-density gate:

$$n_i = |\{j \neq i : \|\mathbf{r}_i - \mathbf{r}_j\| < r_{\text{nbr}}\}| < n_{\text{max}}$$

If the gate passes, a daughter cell is placed at $\mathbf{r}_{\{i'\}} = \mathbf{r}_i + 0.3 \cdot \sigma \cdot (\cos \psi, \sin \psi)$, $\psi \sim U[0, 2\pi)$, with a uniformly random heading. This is a soft Allee / carrying-capacity effect: tumor cells deep inside the colony cannot divide because they are surrounded by neighbors.

3. T-cell dynamics

T cells are fast active Brownian particles with chemotaxis. They are coupled to both scalar fields:

$$\frac{d\mathbf{R}_j}{dt} = v_I \mathbf{m}_j + \chi_a \nabla c_a(\mathbf{R}_j) - \alpha \nabla c_s(\mathbf{R}_j) + \dots + \sqrt{2D_I} \boldsymbol{\zeta}_j(t)$$

where the dots are repulsion sums (T-T and T-tumor) identical in form to the tumor equation, and $\mathbf{m}_j = (\cos \phi_j, \sin \phi_j)$. Heading evolves by:

$$\frac{d\phi_j}{dt} = \sqrt{2D_R^I} \mu_j(t)$$

Chemotaxis: the two key terms

- $+\chi_a \nabla c_a$ pulls T cells UP the attractant gradient, toward tumor signals.
- $-\alpha \nabla c_s$ pushes T cells DOWN the suppressant gradient, away from the tumor halo.

The coupling $\alpha \equiv \chi_s$ is the immunosuppression strength and is ONE of the two control parameters of the phase diagram. The other is $\rho_I \equiv N_I(0)$.

Why this generates phases

Three rates compete at any (ρ_I, α) :

- proliferation rate $\sim p_{div} \cdot N_T$
- T-cell recruitment $\sim \chi_a \times \text{tumor } c_a \text{ flux}$ (grows with N_T)
- T-cell exclusion $\sim \alpha \times \text{tumor } c_s \text{ flux}$ (also grows with N_T)

Different (ρ_I, α) coordinates put different rates on top.

4. Reaction-diffusion fields

Both fields obey a diffusion-source-decay PDE with tumor density ρ_T as the source:

$$\frac{\partial c_a}{\partial t} = D_a \nabla^2 c_a + s_a \rho_T(\mathbf{x}, t) - \lambda_a c_a$$

$$\frac{\partial c_s}{\partial t} = D_s \nabla^2 c_s + s_s \rho_T(\mathbf{x}, t) - \lambda_s c_s$$

The asymmetry that drives the physics

$D_a \gg D_s$ (we use 5 vs 0.5). Attractant diffuses $\sim 3\times$ further than suppressant before decaying:

$$\ell_a = \sqrt{D_a/\lambda_a} \approx 7\sigma, \quad \ell_s = \sqrt{D_s/\lambda_s} \approx 2.2\sigma$$

So T cells can SENSE the tumor from across the box (long-range c_a), but only feel REPULSION close in (short-range c_s). The competition between those scales — with opposite signs in the T-cell equation — is what generates the phase diagram.

Cloud-in-cell deposition

Each tumor cell at grid coords (g_x, g_y) with integer parts (l_x, l_y) and fractional parts (f_x, f_y) contributes weight $w_{\{p,q\}} / (\Delta x)^2$ to corner (l_x+p, l_y+q) for $p, q \in \{0,1\}$, where $w_{\{p,q\}} = (1-f_x)^{1-p} f_x^p \cdot (1-f_y)^{1-q} f_y^q$. Total mass is preserved exactly: $\int \rho_T dA = N_T$.

5. Killing rule & numerical methods

5.1 Killing

For each alive T cell j :

1. Find its nearest alive tumor cell $i^*(j)$ under minimum-image distance.
2. If $\|r_j - r_{i^*}\| < r_{\text{kill}}$ AND $U \sim U[0,1) < p_{\text{kill}}$, remove tumor cell i^* .

Iteration is over T CELLS, not tumor cells, so kill rate scales with N_I (per spec).
At most one kill per T cell per step (no super-killers in dense regions).

5.2 Euler-Maruyama (particles)

$$\mathbf{r}_i^{n+1} = \mathbf{r}_i^n + \Delta t (v_T \mathbf{n}_i + \mathbf{F}_i) + \sqrt{2D_T \Delta t} \boldsymbol{\xi}_i^n$$

$\xi_i^n \sim N(0, I_2)$ i.i.d. across steps and particles. Positions wrapped mod L .

5.3 FTCS Euler + auto-substep (fields)

Five-point stencil; the CFL ratio $\nu = D_{\text{max}} \cdot \Delta t / (\Delta x)^2$ must be $\leq 1/2$. We aim for $\nu_{\text{sub}} \leq 1/4$ and subcycle:

$$n_{\text{sub}} = \max(1, \lceil 4\nu \rceil), \quad \Delta t_{\text{sub}} = \Delta t / n_{\text{sub}}$$

5.4 Performance

Single @njit(cache=True, fastmath=True) inner step. Naive $O(N^2)$ pairwise force — at $N \leq 1500$ this is ~ 1 ms/step on one core. joblib.Parallel for the 192-run sweep: ~ 10 min wall time on 16 cores.

6. Parameter table (production defaults)

Symbol	Code name	Value	Meaning
L	L	100	box side length
G	G	64	field grid resolution
Δt	dt	0.01	timestep
T_f	T_final	100	total simulated time
v_T	v_T	0.1	tumor self-propulsion
D_R^T	D_R_T	0.1	tumor rotational diffusion
σ	sigma_T	1.0	repulsion cutoff (cell diameter)
k_rep	k_rep_T	30	repulsion stiffness
p_div	p_div	0.004	per-step division probability
r_nbr	nbr_radius	1.5	neighbor-count radius (density gate)
n_max	nbr_threshold	6	max neighbors allowed for division
v_I	v_I	1.0	T-cell self-propulsion
D_R^I	D_R_I	1.0	T-cell rotational diffusion
χ_a	chi_a	20.0	attractant chemotactic coupling
$\alpha = \chi_s$	chi_s	scanned	suppressant chemotactic coupling
D_a, D_s	D_a, D_s	5.0, 0.5	field diffusivities
s_a, s_s	s_a, s_s	1.0, 1.0	field source rates
λ_a, λ_s	lam_a, lam_s	0.1, 0.1	field decay rates
r_kill	r_kill	1.5	killing engagement radius
p_kill	p_kill	0.12	per-T-cell per-step kill probability
N_T(0)	N_T_initial	50	initial tumor seed
$\rho_I = N_I(0)$	N_I_initial	scanned	initial T-cell count

Five parameters were tuned away from spec defaults so the three-phase structure becomes visible in the 8-hour time budget: χ_a 5→20, p_kill 0.05→0.12, p_div 0.005→0.004, T_f 200→100, N_T_max 4000→800. See docs/DECISIONS.md.

7. Observables & dimensionless groups

7.1 Primary order parameter

$$\Phi(\rho_I, \alpha) = \text{clip}(N_T(T_f) / N_T(0), 10^{-2}, 10^{+2})$$

Geometrically averaged over S seeds (sensible because order parameter spans orders of magnitude):

$$\bar{\Phi}(\rho_I, \alpha) = \exp\left(S^{-1} \sum_{s=1}^S \log \Phi^{(s)}\right) !$$

7.2 Secondary observables

- $N_T(t)$, $N_I(t)$ trajectories per run
- visual: positions, suppressant field heatmap, T-cell glow → see outputs/videos/

These distinguish:

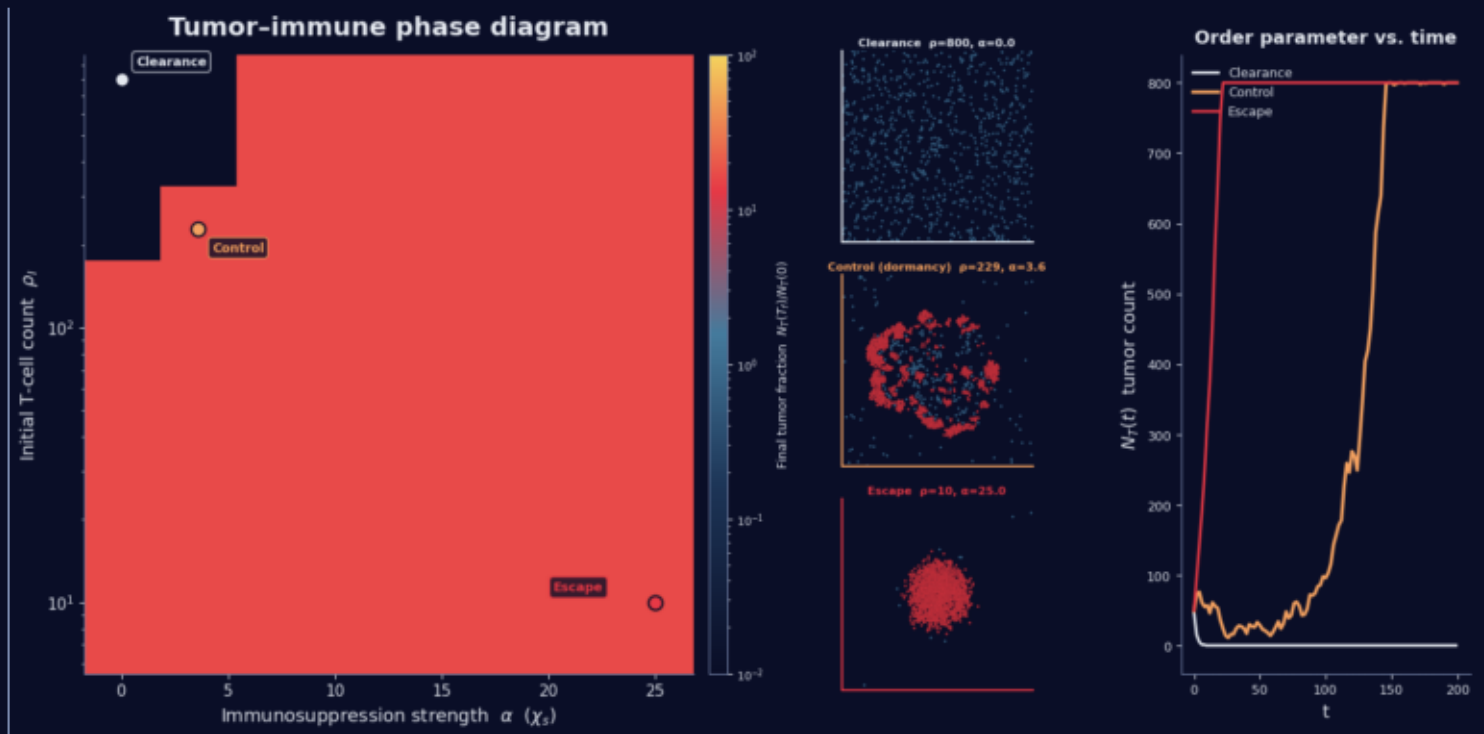
- clearance – monotone fast decay to 0
- control / dormancy – long plateau at intermediate N_T (sometimes followed by late escape)
- escape – monotone fast rise to the array cap

7.3 Dimensionless groups

- Péclet (each species): $Pe_T = Pe_I = v / (D_R \sigma) = 1$ (persistence $\approx$ one cell diameter)
- Attractant range: $\lambda_a = \sqrt{(D_a/\lambda_a)} \approx 7 \sigma$
- Suppressant range: $\lambda_s = \sqrt{(D_s/\lambda_s)} \approx 2.2 \sigma$
- Range ratio: $\lambda_a / \lambda_s \approx 3.2$ – the asymmetry that creates phases
- Suppression ratio: α / χ_a – scanned from 0 to 1.25

8. Phase diagram (numerical result)

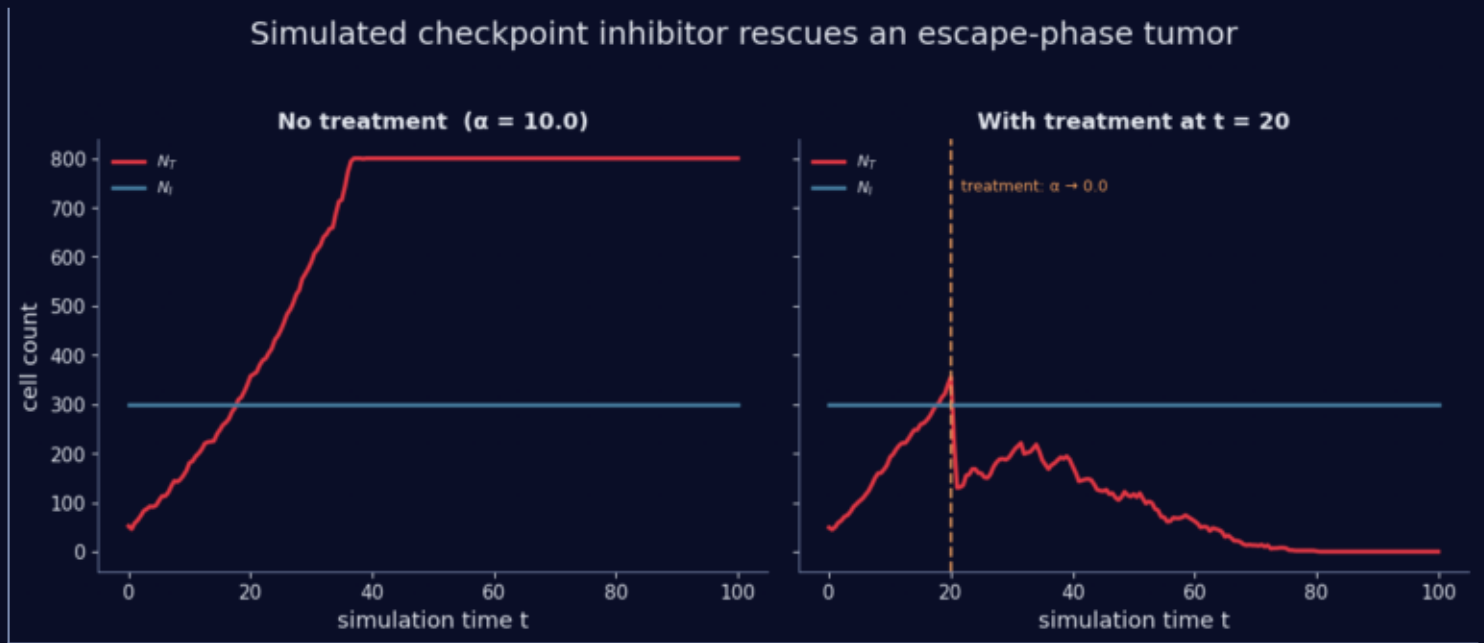
192 runs (8×8 grid $\times$ 3 seeds), final tumor fraction Φ on a log color scale.



Clearance occupies a small wedge at high ρ_I and low α ; almost everywhere else the tumor saturates the array cap (escape). The transition between the two is sharp — that boundary IS the control / dormancy region. Mechanistically this predicts that checkpoint inhibitors should have a THRESHOLD response rather than a smooth dose curve.

9. Treatment experiment (numerical result)

Identical initial state and seed. Treated run flips α : $10 \rightarrow 0$ at $t = 20$.



Headline result

The same initial state goes to two opposite outcomes depending on a single mid-run parameter change. This is exactly the mechanism by which anti-PD-1, anti-PD-L1, and anti-CTLA-4 antibodies rescue immune-excluded tumors clinically.

In our model, lowering α moves the system across the phase boundary in the (ρ_I, α) plane from escape into clearance. No new parameters, no extra physics — just a single switch.